Price regression

Held-out test set: 9,669 listings. Three-fold CV uses training rows only.

Model	MAE (k AZN)	RMSE (k AZN)	R^2	CV log-RMSE
Our tree	103.3	759.1	0.227	0.409 ± 0.018
sklearn tree	103.5	759.2	0.227	0.408 ± 0.018

Luxury tail: 85 listings above 2.5M AZN have 3.64M AZN MAE

The other 9,584 listings have 72.0k AZN MAE. The tree underpredicts 90.6% of the luxury tail.

Premium-tier classification

Premium means price above the 210,000 AZN training median.

Model	Test F1	Test AUC	3-fold CV F1
Our tree	0.904	0.967	0.897 ± 0.001
sklearn tree	0.904	0.967	0.897 ± 0.001
Our linear Pegasos	0.777	0.826	0.774 ± 0.013
sklearn SGD hinge	0.778	0.839	0.778 ± 0.000
Our RFF Pegasos	0.806	0.874	0.807 ± 0.006

The tree leads on F1. RFF raises SVM F1 from 0.777 to 0.806.

Same held-out rows for all models; hyperparameters selected on validation.